

PREDICTING FOOTBALL PLAYERS MARKET VALUE

DAVID ASLANYAN | SUPERVISOR: ARMAN ASRYAN PHD
AKIAN COLLEGE OF SCIENCE AND ENGINEERING | AMERICAN UNIVERSITY OF ARMENIA



Abstract

Transfer values are central to recruitment, squad planning, and contract negotiation, but the link between on-field performance, contract status, and market value is not always transparent. This project builds a reproducible prediction pipeline using season-level player analytics from the top five European leagues over 2021/22–2023/24. The model target is $\log_{10}(\text{market_value_eur})$, which reduces skew while preserving interpretability. With a 300-minute filter, the dataset contains 5,658 player-season rows, split into 3,818 training rows and 1,840 test rows. The best global model is histogram gradient boosting, achieving test RMSE log 0.888, MAE log 0.666, and R^2 0.615. The results show that market value is predictable from observed performance and contract features, though valuation is also shaped by unobserved reputation, club strategy, and market context.

Introduction

Football transfer values reflect both sporting performance and economic opportunity, but their drivers remain partially opaque. Publicly available player statistics now allow systematic analysis of how performance, age, position, league, and contract expiry combine to determine market value. Many existing valuation models focus on a narrow feature set, specific leagues, or predictive accuracy without sufficient interpretability. This project expands the analysis by: modeling transfer value from season-level performance and contract data, evaluating positional differences across defenders, midfielders, and forwards, analyzing whether league-specific or position-specific models improve prediction, and examining how contract years remaining influences market value after controlling for performance.

Data Source: season-level observations from Premier League, La Liga, Serie A, Bundesliga, and Ligue 1 seasons: 2021/22, 2022/23, 2023/24

Features included:

- age, age²
- minutes, starts, 90s
- goals_per90, assists_per90, shots_per90, shots_on_target_per90
- crosses_per90, interceptions_per90, tackles_won_per90
- fouls_per90, fouled_per90
- contract_years_remaining, contract_missing
- days_since_last_transfer, transfer_count_before_valuation, loan_count_before_valuation
- position_group, cleaned_comp, season

Dataset construction:

- target: $\log_{10}(\text{market_value_eur}) = \log_{10}(\text{market_value_eur})$
- rows with fewer than 300 minutes excluded
- final dataset: 5,658 rows
- training: 3,818 rows from 21/22 + 22/23
- test: 1,840 rows from 23/24

Results

Key model performance

- Best model: hist_gradient_boosting
- Test RMSE log: 0.8877
- Test MAE log: 0.6663
- Test R^2 : 0.6146
- Test MAE EUR: €6,871,178

Models tested:

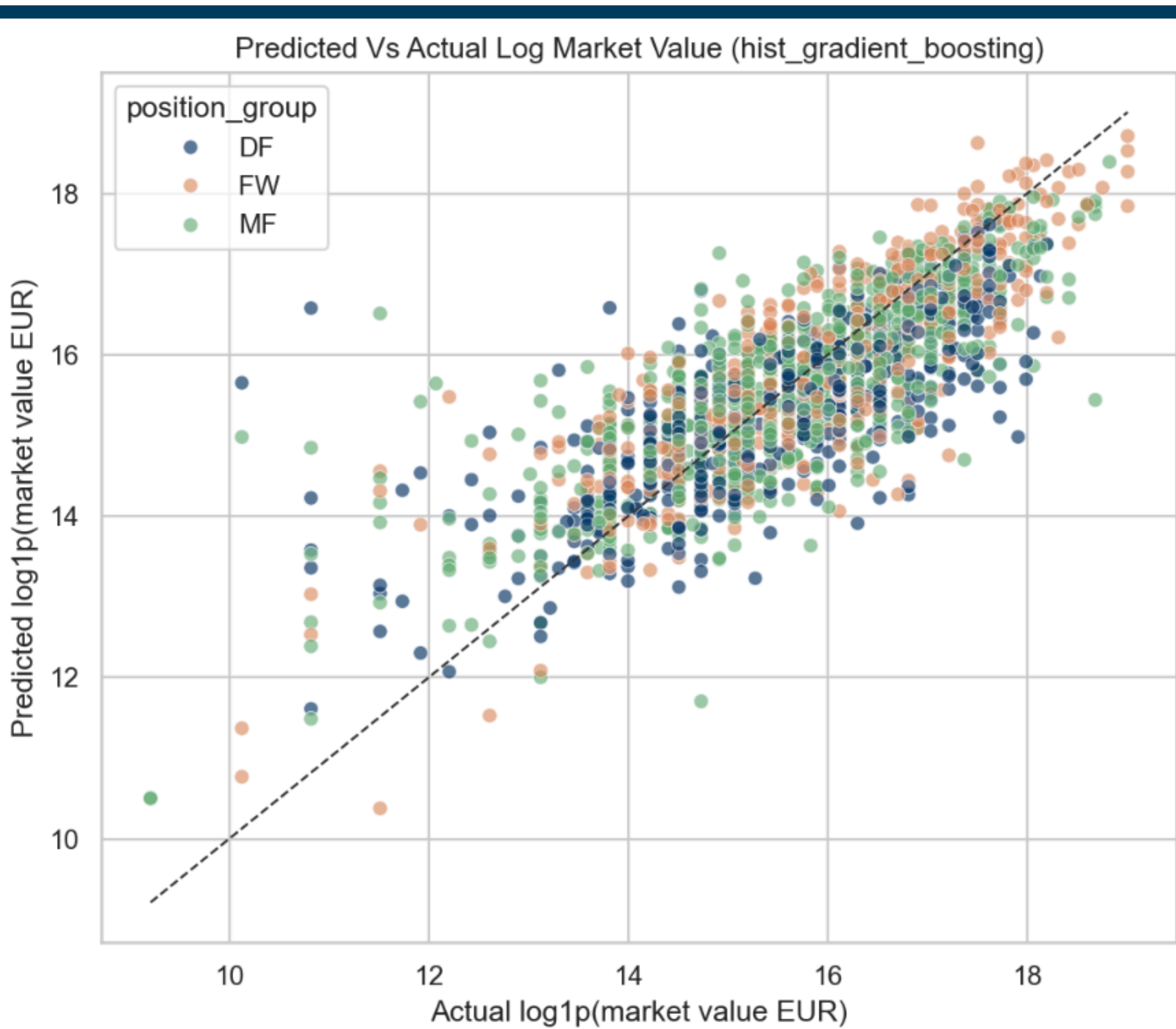
- mean baseline
- linear regression
- ridge regression
- histogram gradient boosting

Evaluation:

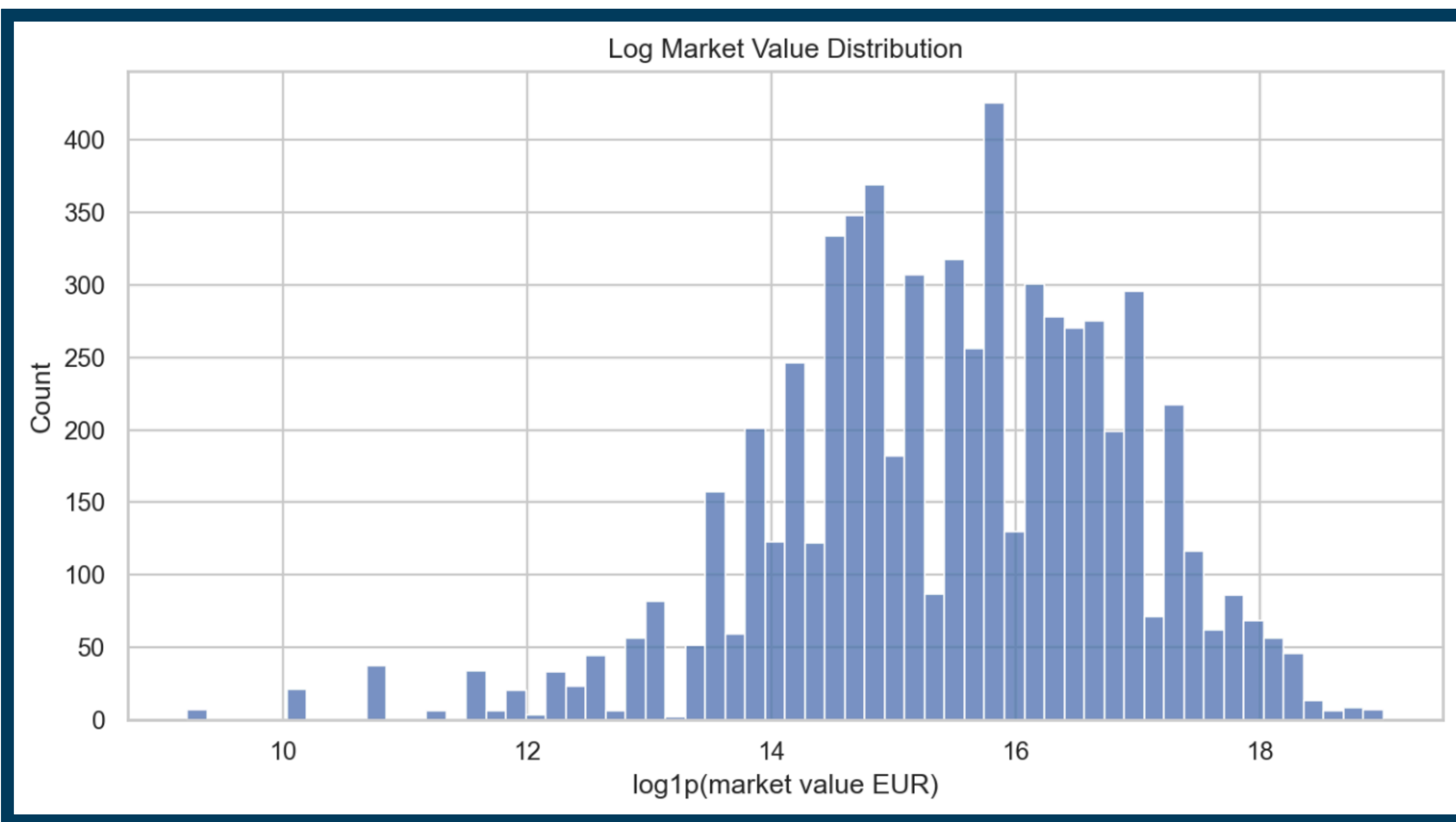
- holdout test season (2023/24)
- metrics: RMSE log, MAE log, R^2 , MAE EUR
- global model vs position-specific and league-specific models

model	rows	rmse_log	mae_log	r2	mae_eur
hist_gradient_boosting	1840	0.8877	0.6663	0.6146	6871178.1624
ridge_alpha_1	1840	0.9925	0.7341	0.5183	7665130.2120
linear_regression	1840	0.9925	0.7341	0.5183	7668460.5645
mean_baseline	1840	1.4320	1.1333	-0.0027	11046357.1685

- Histogram gradient boosting outperforms ridge, linear regression, and the mean baseline on the 2023/24 holdout.
- This validates a nonlinear approach over simple linear models for market value prediction.



Predictions cluster near the identity line, demonstrating good calibration across value levels.



The log-transformed market value distribution is much more balanced than raw value. This confirms that $\log_{10}$ is appropriate for regression.

Conclusion

- A reproducible global model can estimate football player market value from season-level performance and contract data.
- Histogram gradient boosting is the strongest model, achieving the best held-out accuracy.
- League and position are important value signals, but the global model generalizes better than specialized segmentation.
- Key valuation drivers include:
 - age,
 - minutes played,
 - contract years remaining,
 - league,
 - position.
- Transfer values are still partly influenced by unobserved factors such as reputation, club strategy, and market dynamics.

References

- Transfermarkt valuation literature and market value proxy studies
- Sports economics and football analytics on player valuation
- Machine learning regression and ensemble modeling research for predictive analytics